AI 623 - Deep Vision Language Models (Introduction)

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Abstract

This report establishes experimental and conceptual foundations for several core architectures used in modern deep vision and multimodal learning. Specifically, we analyze the internal mechanisms of ResNet-152, investigate attention-based representations in Vision Transformers (ViT), train and evaluate Variational Autoencoders (VAE), and study cross-modal alignment in CLIP. Emphasis is placed on reproducible experimentation, architectural design choices, and empirical analysis through visualization and controlled ablations. Together, these experiments provide practical insight into representation learning, transferability, generative modeling, and vision–language alignment.

Code Repository: https://github.com/nausherwan-malik/Deep-Vision-Language-Models.git

1. Introduction

Recent advances in deep learning have been driven by architectures that prioritize scalability, robust representation learning, and multimodal integration. Convolutional neural networks such as Residual Networks (ResNet) mitigate optimization difficulties in very deep models through identity skip connections, enabling stable gradient propagation and strong transfer learning performance. In contrast, Vision Transformers (ViT) adopt a self-attention-based formulation that models global context across image patches, offering improved flexibility and interpretability through explicit attention mechanisms.

Beyond discriminative modeling, generative frameworks such as Variational Autoencoders (VAE) provide a principled approach to unsupervised learning by enforcing structured latent representations via probabilistic regularization. These models enable controlled generation and facilitate analysis of latent space organization. At the multimodal level, Contrastive Language–Image Pretraining (CLIP) extends representation learning across vision and language by jointly embedding images and text into a shared semantic space, enabling zero-shot classification and cross-modal retrieval without task-specific supervision.

This report experimentally examines these architectures in the context of deep vision-language systems. Through controlled training, feature visualization, attention analysis, and embedding alignment, we aim to characterize the practical behavior of each model class and highlight the trade-offs between architectural inductive biases, computational efficiency, and representational capacity.

2. Residual Networks (ResNet-152)

2.1. Motivation and Architecture

Deep convolutional networks often suffer from optimization difficulties as depth increases, leading to vanishing gradients and degraded training performance. Residual Networks address this limitation by introducing identity skip connections that allow layers to learn residual mappings rather than direct transformations. Formally, a residual block computes

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, W) + \mathbf{x}, \tag{1}$$

where $\mathcal{F}$ denotes the residual function and $\mathbf{x}$ is the block input. This formulation enables more stable gradient propagation and facilitates the training of very deep architectures such as ResNet-152.

ResNet-152 consists of stacked bottleneck residual blocks that combine 1×1, 3×3, and 1×1 convolutions to reduce computational cost while preserving representational capacity. The pretrained model used in this work was originally optimized on ImageNet and serves as a strong backbone for downstream transfer learning tasks.

2.2. Transfer Learning Setup

To evaluate feature transferability, the pretrained ResNet-152 backbone was adapted to CIFAR-10 by replacing the final classification layer and freezing all convolutional layers. Only the linear classification head was trained, allowing us to isolate the quality of pretrained visual representations while minimizing computational overhead. Training and validation performance were recorded across multiple epochs.

2.3. Baseline Transfer Learning Results

Table 1 reports classification accuracy across three training epochs. Despite freezing over 99% of model parameters, the network achieved strong validation performance and converged rapidly.

Table 1. Baseline ResNet-152 transfer learning performance with frozen backbone.

Epoch	Train Accuracy (%)	Validation Accuracy (%)
1	78.44	82.58
2	83.54	84.63
3	84.80	84.56

The strong validation accuracy demonstrates that pretrained convolutional features generalize effectively to smaller-scale datasets. This result highlights the practicality of transfer learning for reducing training cost while maintaining competitive performance.

2.4. Effect of Residual Connections

To analyze the functional role of skip connections, identity shortcuts were disabled in the first two bottleneck blocks of layer3. The modified network was retrained under the same experimental conditions as the baseline.

As shown in Table 2, removing residual connections resulted in severe degradation of both training and validation accuracy.

Table 2. Performance after disabling residual skip connections.

Epoch	Train Acc. (%)	Val Acc. (%)
1	16.22	18.17
2	18.66	18.14
3	18.98	19.41

The network failed to converge beyond near-random performance, confirming that residual connections are critical for stable optimization and effective gradient propagation in deep architectures.

2.5. Feature Hierarchy Visualization

Intermediate feature representations were extracted from early (layer1), middle (layer3), and late (layer4) layers using forward hooks. Global average pooling was applied to each feature map before dimensionality reduction with t-SNE.

Figure 1 visualizes the resulting embeddings. Early-layer features exhibit significant class overlap, reflecting low-level texture representations. Middle-layer features show partial clustering, while late-layer representations demonstrate improved class separability and compact semantic groupings. This progression illustrates how hierarchical feature abstraction emerges across network depth.

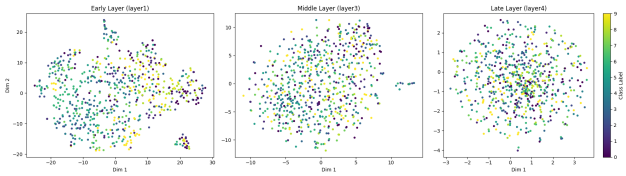

Figure 1. t-SNE projections of ResNet-152 feature representations extracted from early, middle, and late layers.

2.6. Fine-Tuning Strategy Comparison

To analyze the impact of pretrained initialization and fine-tuning scope, three training configurations were evaluated: pretrained partial fine-tuning, pretrained full fine-tuning, and full fine-tuning from random initialization. Partial fine-tuning updates only the final residual block and classification head, whereas full fine-tuning updates all network parameters.

Table 3 summarizes the final epoch performance for each setting.

Table 3. Comparison of fine-tuning strategies on CIFAR-10 using ResNet-152.

Training Strategy	Train Acc. (%)	Val Acc. (%)
Pretrained + Partial FT	98.93	90.58
Pretrained + Full FT	99.18	96.77
Scratch + Full FT	69.89	68.60

Full fine-tuning with pretrained weights achieved the highest validation accuracy, indicating that adapting all layers enables better specialization to the target dataset. Partial fine-tuning achieved competitive performance while updating substantially fewer parameters, offering a favorable trade-off between computational efficiency and accuracy.

In contrast, training from random initialization converged slowly and resulted in significantly lower validation performance, demonstrating the strong dependence of deep convolutional networks on large-scale pretraining and highlighting the inefficiency of training very deep architectures from scratch on limited datasets.

3. Vision Transformers (ViT)

3.1. Motivation and Architecture Overview

Vision Transformers model images as sequences of fixed-size patches and replace convolutional feature extraction with self-attention-based token interactions. Each image is decomposed into non-overlapping patches, which are

linearly embedded and combined with positional encodings before being processed by stacked transformer encoder blocks. A special classification token (CLS) is prepended to the patch sequence and serves as the global image representation used for prediction.

Unlike convolutional networks that rely on local receptive fields and hierarchical feature extraction, ViTs leverage global self-attention at every layer. This design enables long-range dependency modeling and provides direct access to attention weights, which can be used for interpretability and analysis.

3.2. Zero-Shot Image Classification

A pretrained ViT model (`vit-base-patch16-224`) was evaluated on two natural images without additional fine-tuning. For the input image `car.jpg`, the model predicted the class *convertible*, while for `cat.jpg`, the predicted label was *tabby cat*. In both cases, the predicted classes accurately reflected the dominant objects present in the images.

These results demonstrate the strong zero-shot generalization capability of pretrained Vision Transformers, which leverage large-scale pretraining to produce semantically meaningful predictions on previously unseen samples.

Figure 2. CLS-token attention overlay for the car image. High-intensity regions indicate patches receiving stronger attention during classification.

3.3. Patch Attention Visualization

To analyze spatial attention patterns, attention matrices from the final transformer layer were extracted and averaged across attention heads. The attention weights corresponding to the classification token were reshaped into a two-dimensional patch grid and overlaid on the original images.

Figure 2 and Figure 3 show representative attention overlays for the car and cat images, respectively. For the car image, attention is concentrated around the front region. For the cat image, attention primarily focuses on the face, which are semantically informative regions for classification. We also inspected individual attention heads in the final layer and observed moderate specialization, with some heads focusing on object boundaries while others emphasized global context. This behavior suggests complementary roles across heads in aggregating spatial and semantic cues.

Figure 3. CLS-token attention overlay for the cat image. The model focuses primarily on the face and upper body regions.

3.4. Multi-Layer Attention Aggregation

To reduce patch-level noise and improve spatial consistency, attention maps from the final four transformer layers were aggregated by averaging across layers and attention heads. This smoothing operation produced more coherent attention distributions and reduced fragmentation across neighboring patches.

Fig 4 represents the multi-layer attention of last k layers

for the image corresponding to the tabby cat. Compared to single-layer attention maps, the aggregated representations exhibited improved alignment with object boundaries and more stable activation regions, indicating that deeper transformer layers encode increasingly refined semantic attention patterns.

Figure 4. Multi-layer aggregated CLS-token attention map for the tabby cat image using the last k transformer layers. Aggregation improves spatial coherence and semantic focus.

Figure 5. Effect of random and center-based patch masking on ViT predictions for the car image.

Figure 6. Effect of random and center-based patch masking on ViT predictions for the cat image.

3.5. Robustness to Patch Masking

The robustness of Vision Transformer predictions was evaluated by masking different fractions of input patches during inference. Two masking strategies were considered: random patch masking and structured center-region masking, with mask ratios of 0.3 and 0.6.

Figure 5 and Figure 6 illustrate representative masking results for the car and cat images. Under moderate random masking (30%), the model retained correct predictions with only a slight reduction in confidence. In contrast, center masking caused larger confidence drops and more frequent changes in predicted class labels. At higher mask ratios (60%), both masking strategies significantly degraded prediction confidence, with structured occlusion producing the strongest performance deterioration.

These observations suggest that Vision Transformers leverage distributed contextual information across patches, enabling partial robustness to random occlusions, while remaining sensitive to the removal of spatially critical semantic regions.

4. Variational Autoencoders (VAE)

4.1. Model Architecture and Training Setup

A Variational Autoencoder was trained on the FashionMNIST dataset to study unsupervised representation learning and generative modeling. The model consists of an encoder network that maps input images to a latent distribution parameterized by mean and log-variance vectors, and a decoder network that reconstructs images from sampled latent variables. A latent dimensionality of 20 was used in all experiments.

The training objective optimizes the Evidence Lower Bound (ELBO), composed of a reconstruction term and a Kullback–Leibler (KL) divergence regularization term. Mean squared error (MSE) was used for reconstruction loss, while the KL divergence enforces alignment between the learned posterior distribution and the standard normal prior.

To mitigate posterior collapse during early training, KL warm-up was applied by linearly annealing the KL weight over the first three epochs. The model was trained for five epochs using the Adam optimizer with a learning rate of 10^{-3}.

4.2. Training and Validation Performance

Figure 7 shows the training and validation loss curves over epochs. Both losses decrease and converge to similar values, indicating stable optimization and limited overfitting.

After five epochs, the final training and validation losses reached 24.72 and 24.78 respectively, demonstrating consistent generalization behavior across the dataset.

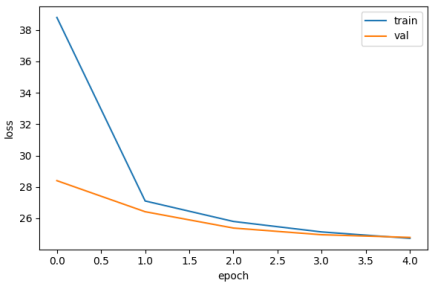

Figure 7. Training and validation ELBO loss curves for the VAE model on FashionMNIST.

4.3. Reconstruction Quality

To evaluate reconstruction performance, original validation images were passed through the encoder-decoder pipeline. Figure 8 shows representative reconstruction examples.

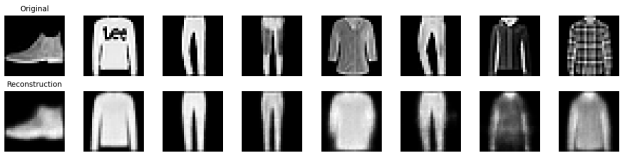

Figure 8. Original FashionMNIST images (top row) and corresponding VAE reconstructions (bottom row).

The reconstructed images preserve overall object structure and silhouette while exhibiting slight blurring, which is characteristic of VAEs trained with pixel-wise reconstruction objectives.

4.4. Generative Sampling from Different Priors

To analyze generative behavior, samples were drawn from both Gaussian and Laplacian latent priors and decoded into image space.

Figure 9 shows samples generated using the standard Gaussian prior, while Figure 10 displays samples obtained from a Laplacian prior.

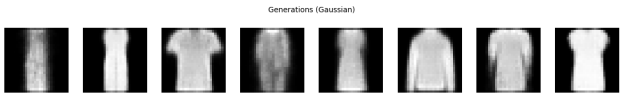

Figure 9. Samples generated from the decoder using a Gaussian latent prior.

Gaussian sampling produced smoother and more consistent outputs, while Laplacian sampling resulted in higher contrast variations and occasionally sharper edges. These differences reflect the influence of latent prior distributions on generative diversity and output characteristics.

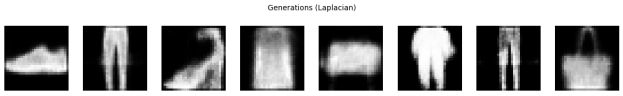

Figure 10. Samples generated from the decoder using a Laplacian latent prior.

4.5. Posterior Collapse Analysis

To evaluate whether posterior collapse occurred, ELBO components were analyzed on the validation set. The average reconstruction loss per sample was 17.28, while the average KL divergence per sample was 7.48. The mean latent distribution statistics were $\mu \approx 0.0077$ and $\log \sigma^2 \approx -0.74$.

The non-trivial KL magnitude indicates that the latent variables retain meaningful information about the input data rather than collapsing entirely to the prior distribution. This suggests that KL warm-up successfully mitigated posterior collapse by allowing the encoder to learn informative representations before full regularization was applied.

Overall, these results demonstrate stable VAE training behaviour, effective latent space utilization, and successful generative sampling across different prior distributions.

5. CLIP: Modality Gap Analysis

5.1. Zero-Shot Classification on STL-10

CLIP was evaluated on the STL-10 test set using zero-shot classification with three prompt strategies: plain class labels, photo-style prompts, and descriptive prompts. Image and text embeddings were normalized and compared using cosine similarity.

Table 4 summarizes the resulting accuracies.

Table 4. Zero-shot classification accuracy of CLIP on STL-10 under different prompt formulations.

Prompt Strategy	Accuracy (%)
Plain labels	96.24
Photo prompt	97.38
Descriptive prompt	97.44

Prompt engineering significantly improved classification performance. Adding natural language context such as *"a photo of a {class}"* increased accuracy by more than 1% compared to plain labels, demonstrating CLIP's sensitivity to linguistic structure and semantic context.

5.2. Exploring the Vision–Language Modality Gap

To visualize the embedding distributions, image and text features were projected into two dimensions using UMAP. Two settings were compared: raw embeddings and L2-

normalized embeddings.

Figure 11 shows the resulting projections. Without normalization, image and text embeddings occupy partially separated regions of the embedding space. After normalization, the two modalities exhibit improved overlap and tighter clustering, indicating better alignment under cosine similarity.

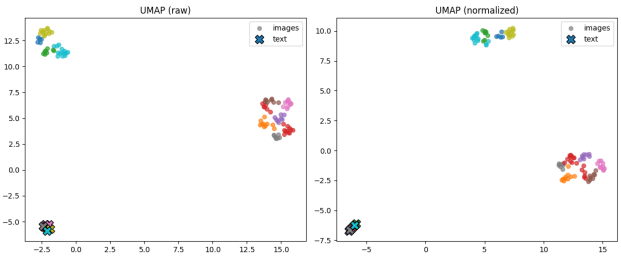

Figure 11. UMAP visualization of CLIP image and text embeddings before and after feature normalization. Normalization improves cross-modal alignment and cluster overlap.

Despite the presence of a visible modality gap, CLIP achieves strong classification performance because similarity-based retrieval relies on relative distances rather than absolute embedding overlap.

5.3. Bridging the Modality Gap via Procrustes Alignment

To further reduce the modality gap, orthogonal Procrustes alignment was applied to rotate image embeddings toward their corresponding class text embeddings. Each image embedding was paired with the text embedding of its ground-truth class label to learn an optimal rotation matrix.

Figure 12 visualizes the embedding distributions before and after alignment. After applying the rotation transform, image and text embeddings exhibit stronger spatial overlap and improved class-wise grouping.

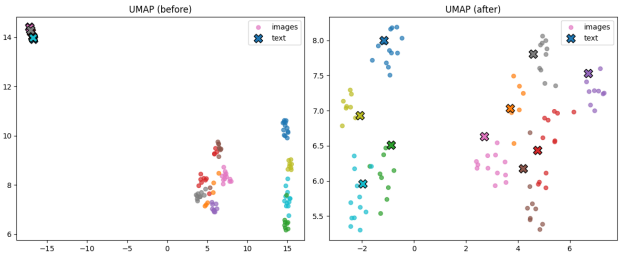

Figure 12. UMAP visualization of CLIP embeddings before and after Procrustes alignment. Alignment improves cross-modal clustering consistency.

5.4. Effect of Alignment on Classification Performance

The impact of alignment was evaluated by recomputing zero-shot classification accuracy on the STL-10 test set.

Table 5. Classification accuracy before and after Procrustes alignment.

Embedding Configuration	Accuracy (%)
Before alignment	97.38
After alignment	97.78

The aligned embeddings produced a modest but consistent improvement in accuracy. This result suggests that while CLIP embeddings are already well-structured, simple linear alignment techniques can further enhance cross-modal consistency without retraining the model.

Overall, these experiments demonstrate that CLIP effectively bridges vision and language representations through contrastive pretraining, while still exhibiting measurable modality gaps that can be partially mitigated through post-hoc alignment.